

Quantum hydrogen–bond symmetrization in the superconducting hydrogen sulfide system

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The quantum nature of the proton can crucially affect the structural and physical properties of hydrogen compounds. For example, in the high-pressure phases^{1,2} of H₂O, quantum proton fluctuations lead to symmetrization of the hydrogen bond and reduce the boundary between asymmetric and symmetric structures in the phase diagram by 30 gigapascals (ref. 3). Here we show that an analogous quantum symmetrization occurs in the recently discovered⁴ sulfur hydride superconductor with a superconducting transition temperature T_c of 203 kelvin at 155 gigapascals—the highest T_c reported for any superconductor so far. Superconductivity occurs via the formation of a compound with chemical formula H₃S (sulfur trihydride) with sulfur atoms arranged on a body-centred cubic lattice^{5–9}. If the hydrogen atoms are treated as classical particles, then for pressures greater than about 175 gigapascals they are predicted to sit exactly halfway between two sulfur atoms in a structure with $Im\bar{3}m$ symmetry. At lower pressures, the hydrogen atoms move to an off-centre position, forming a short H–S covalent bond and a longer H...S hydrogen bond in a structure with $R3m$ symmetry^{5–9}. X-ray diffraction experiments confirm the H₃S stoichiometry and the sulfur lattice sites, but were unable to discriminate between the two phases¹⁰. *Ab initio* density-functional-theory calculations show that quantum nuclear motion lowers the symmetrization pressure by 72 gigapascals for H₃S and by 60 gigapascals for D₃S. Consequently, we predict that the $Im\bar{3}m$ phase dominates the pressure range within which the high T_c was measured. The observed pressure dependence of T_c is accurately reproduced in our calculations for the $Im\bar{3}m$ phase, but not for the $R3m$ phase. Therefore, the quantum nature of the proton fundamentally changes the superconducting phase diagram of H₃S.

The recent discovery of high-temperature superconductivity in compressed hydrogen sulfide⁴ has led to a number of theoretical studies aimed at understanding the phase diagram of the H–S system and the origin of the astonishingly high T_c that was observed^{5–9,12–18}. The overall consensus is that H₂S, the only stable compound formed by hydrogen and sulfur at ambient conditions, is metastable at high pressures and its decomposition gives rise to several H–S compounds. High- T_c superconductivity is believed to occur in a structure with H₃S stoichiometry, and is considered to be conventional in nature, that is, mediated by electron–phonon interactions^{4,5,7,9,12–17}. Alternatives to conventional superconductivity have also been discussed¹⁸. According to structural predictions^{5–9}, H₃S adopts a rhombohedral $R3m$ form between approximately 112 GPa and 175 GPa, and a cubic $Im\bar{3}m$ form at higher pressures. As shown in Fig. 1, the $R3m$ phase is characterized by covalently bonded SH₃ units with a covalent H–S bond of length d_1 . Each of these H atoms is bonded to the next S atom by a hydrogen

H...S bond of length d_2 . By contrast, the $Im\bar{3}m$ phase has full cubic symmetry, with $d_1 = d_2$ so that each H atom resides midway between the two S atoms, as shown in Fig. 1. The $R3m$ structure is nevertheless very close to cubic symmetry; for example, the density functional theory (DFT)-relaxed $R3m$ structure, which represents the minimum of the Born–Oppenheimer energy surface (BOES), has a rhombohedral angle of 109.49° at approximately 150 GPa, compared to 109.47° for a perfect bcc lattice. We verified that imposing a cubic angle on the $R3m$ structure has a negligible effect on the energy difference between the $R3m$ and $Im\bar{3}m$ structures. Consequently, we assume a cubic lattice for both phases in the following.

The bond-symmetrizing second-order transition from $R3m$ to $Im\bar{3}m$ occurs at 175 GPa according to our static lattice calculations. At this pressure, our harmonic phonon calculations show that a Γ -point optical phonon of the high-symmetry $Im\bar{3}m$ phase becomes imaginary, implying that $Im\bar{3}m$ is at a saddle point of the BOES between 112 GPa and 175 GPa, whereas the $R3m$ phase lies at the minimum. Crystal symmetry guarantees that the transition is second-order (see Methods for a symmetry analysis). As occurs in the high-pressure ice X phase^{3,19,20} and other hydrogenated compounds²¹, the quantum nature of the proton can radically alter the pressure at which the second-order phase transition occurs and, in the present case, can strongly affect the stability of the $R3m$ phase below 175 GPa. Determining the stability ranges of these phases therefore requires the inclusion of vibrational zero-point energy (ZPE) along with the static BOES energy. However, the presence of imaginary phonon frequencies

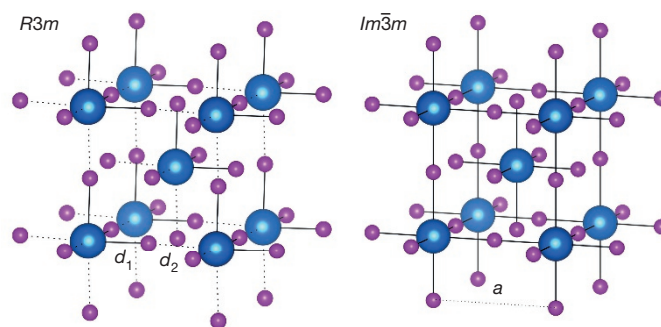


Figure 1 | Crystal structures of the competing phases. Crystal structure in the conventional bcc cell of the $R3m$ (left) and $Im\bar{3}m$ (right) phases. In the $R3m$ structure, the H–S covalent bond of length d_1 is marked with a solid line and the longer H...S hydrogen bond of length d_2 with a dotted line. In the $Im\bar{3}m$ phase, $d_1 = d_2$. The lattice parameter a is marked in the $Im\bar{3}m$ structure. Blue and pink atoms represent S and H atoms, respectively.

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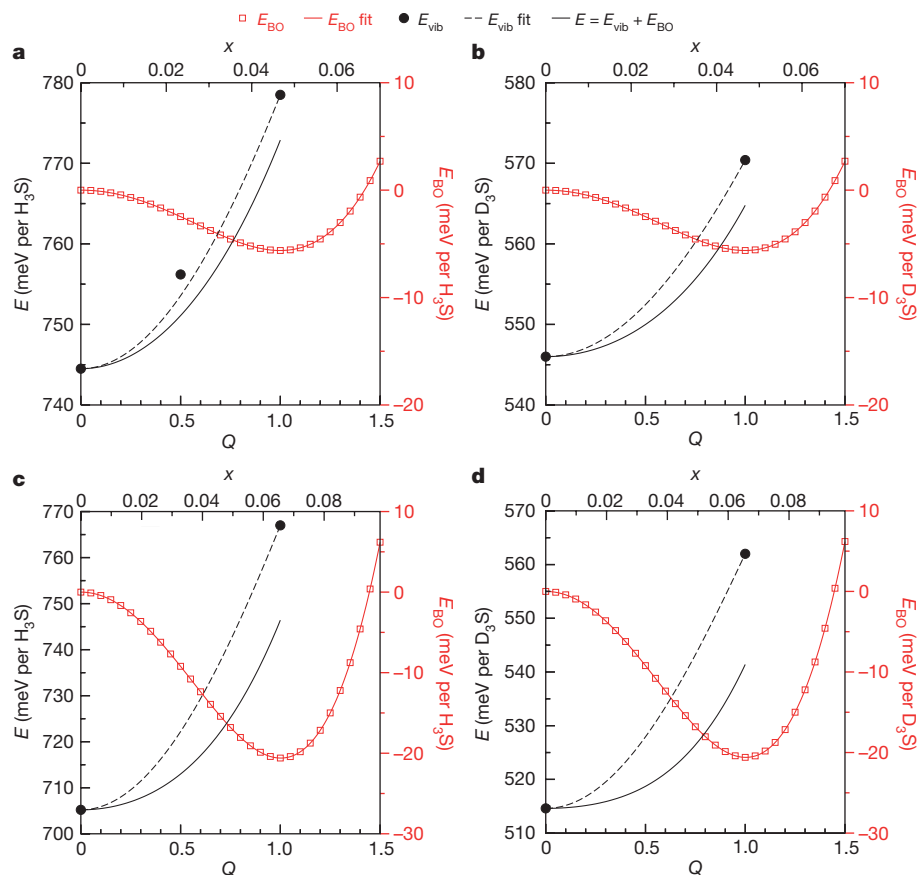


Figure 2 | Energetics. **a–d**, E_{BO} , E_{vib} and $E = E_{\text{vib}} + E_{\text{BO}}$ curves for H_2S (**a**, **c**) and D_2S (**b**, **d**) are shown as functions of the reaction coordinate Q that transforms the $Im\bar{3}m$ structure ($Q = 0$) into the $R3m$ structure ($Q = 1$), as well as the relative coordinate $x = (d_2 - a/2)/(a/2)$ that measures the off-centring of the H atoms, where d_2 is the length of the hydrogen bond and a is the lattice parameter (see Fig. 1). The left-hand (black) axes show the energy scale for E_{vib} and E ; the right-hand (red) axes show the scale for E_{BO} . Crystal symmetry implies that $E(Q) = E(-Q)$, so that the curves can be fitted to polynomials with only even terms; this guarantees that the

transition is second-order according to Landau theory¹¹. Results are presented for two different volumes of the primitive bcc lattice; $V = 97.85a_0^3$ (**a**, **b**) corresponds to approximately 150 GPa and $V = 102.11a_0^3$ (**c**, **d**) to 130 GPa. The pressure associated with each volume depends on both the isotope and Q (see Extended Data Fig. 1 for the equations of state). Black circles represent calculated E_{vib} points and the black dashed line the fitted $E_{\text{vib}}(Q)$ curve (see Methods). The $E(Q)$ curve is obtained by addition of the fitted E_{vib} and E_{BO} curves.

hinders calculations of the ZPE, because the quasi-harmonic approximation breaks down, and anharmonicity becomes important.

To elucidate the role of anharmonicity and quantum effects in the pressure range in which the record $T_c = 203$ K was observed, we make use of the stochastic self-consistent harmonic approximation (SSCHA)^{22,23}. The variational SSCHA method was devised for calculating the free energy and phonon spectra while fully incorporating quantum and anharmonic effects, and so is perfectly suited for our purpose. All of the calculations presented here are performed at 0 K. Primitive cells for the $R3m$ and $Im\bar{3}m$ structures contain four atoms (one S atom and three H atoms); consequently, a particular nuclear configuration can be described by a 12-dimensional vector $\mathbf{R}$ containing the atomic coordinates. In the classical limit, the ZPE is neglected and the energy of a nuclear configuration $\mathbf{R}$ is given by the DFT Born–Oppenheimer energy $E_{\text{BO}}(\mathbf{R})$. In the SSCHA, the ZPE is accounted for by approximating the nuclear wavefunction by a Gaussian centred on a centroid coordinate $\mathbf{R}_c$, which denotes the average and most probable position of the nuclei. For a given $\mathbf{R}_c$, the width of the Gaussian is obtained by variational minimization of the expectation value of the sum of the nuclear potential and kinetic energies. In the following analysis, it is convenient to split the SSCHA total energy $E(\mathbf{R}_c)$ into static and anharmonic-vibrational-ZPE contributions: $E(\mathbf{R}_c) = E_{\text{BO}}(\mathbf{R}_c) + E_{\text{vib}}(\mathbf{R}_c)$.

We study the energy landscape $E(\mathbf{R}_c)$ along the line defined by $\mathbf{R}_c(Q) = \mathbf{R}_{Im\bar{3}m} + Q(\mathbf{R}_{R3m} - \mathbf{R}_{Im\bar{3}m})$, in which $\mathbf{R}_{Im\bar{3}m}$ and $\mathbf{R}_{R3m}$ are,

respectively, the coordinates corresponding to the saddle point and minimum of the BOES, representing the two different symmetries. Here, Q is a real number describing the reaction coordinate, so that at $Q = 0$ the centroids are located at the atomic positions of $Im\bar{3}m$, and at $Q = 1$ at the atomic positions of $R3m$. Hence, Q measures the off-centring of the hydrogen nuclear wavefunction and can be associated with the relative coordinate $x = (d_2 - a/2)/(a/2)$ that quantifies the length of the H...S hydrogen bond with respect to the symmetric position (a is the lattice parameter). We analyse the curve $E(\mathbf{R}_c(Q)) \equiv E(Q)$ for a fixed primitive bcc unit cell. As shown in Fig. 2a for a cell volume of $97.85a_0^3$ (a_0 is the Bohr radius), the $E_{\text{BO}}(Q)$ curve has a shallow double-well structure that favours the $R3m$ structure by only 5.6 meV per H_2S . However, after adding the $E_{\text{vib}}(Q)$ energy calculated with the SSCHA, the full $E(Q)$ curve shows a clear minimum at $Q = 0$, which favours the $Im\bar{3}m$ structure. At a larger volume of $102.11a_0^3$, which corresponds to a pressure of about 130 GPa, the minimum of $E(Q)$ is also at $Q = 0$, despite the fact that the one-dimensional Born–Oppenheimer well in $E_{\text{BO}}(Q)$ becomes deeper, as shown in Fig. 2c. Upon repeating these calculations for D_2S , we find that the $Im\bar{3}m$ structure is the most favourable once the ZPE has been included. Therefore, we conclude that the quantum nature of the nuclei symmetrizes the hydrogen bond and leads to a proton wavefunction that is centred at the atomic positions of $Im\bar{3}m$ for both H_2S and D_2S . To eliminate the possibility that the energy minimum occurs beyond the $\mathbf{R}_c(Q)$ line studied, we performed an unconstrained SSCHA

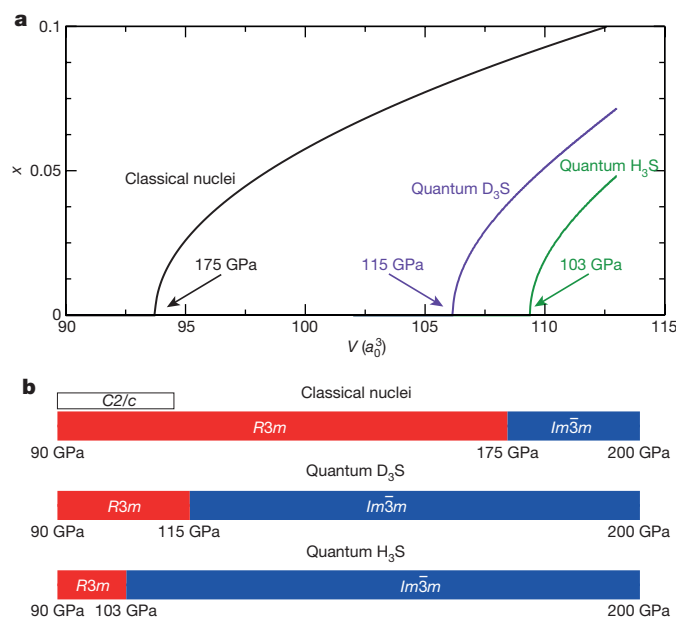


Figure 3 | Second-order phase transition. **a**, For each volume V we plot the relative coordinate x that yields the minimum total energy; x measures the off-centring of the H atoms (see Fig. 2). The results are shown in the classical nuclei limit (black), as well as in the quantum case both for H_3S (green) and D_3S (purple). The volume at which x departs from zero marks the second-order phase transition from the $Im\bar{3}m$ phase to the $R3m$ phase. Transition pressures are also indicated, and include the effects of vibrational energies. **b**, Phase diagram for the second-order phase transition as a function of pressure. As shown in ref. 8, below 112 GPa, H_3S adopts a very different $C2/c$ phase; we mark the expected emergence of this phase by a box.

minimization, optimizing both the width of the Gaussians and the R_c centroid positions. The results of this minimization confirm that, within stochastic error, the centroid position obtained corresponds to

the $Im\bar{3}m$ structure, in which the H–S covalent and H...S hydrogen bond distances equalize, leading to symmetric hydrogen bonds.

The difference between the vibrational energies of $R3m$ and $Im\bar{3}m$ as a function of the x coordinate is weakly dependent on volume. This allows us to interpolate $E(x)$ in a wide volume range and estimate the pressure at which the proton wavefunction shifts away from the centred position. Our calculations show that this symmetry breaking occurs at 103 GPa in H_3S and 115 GPa in D_3S (see Fig. 3). The higher transition pressure in D_3S is due to weaker quantum effects. This isotope effect is similar to that observed in the ice VII/ice X transition²⁴. Considering that, below 112 GPa, the $R3m$ phase is expected to transform into a very different monoclinic $C2/c$ phase consisting of isolated H_2S and H_2 molecules with H_3S stoichiometry⁸, $R3m$ H_3S might not be formed. However, D_3S may adopt the $R3m$ structure at pressures below the transition to the $Im\bar{3}m$ phase.

The quantum proton symmetrization has a very large impact on the phonon spectra of H_3S . As mentioned previously, and shown in Fig. 4, the phonon spectra of $Im\bar{3}m$ H_3S have several imaginary modes in the harmonic approximation below 175 GPa. The corresponding anharmonic SSCHA phonon spectra for $Im\bar{3}m$ H_3S show well-behaved phonon dispersion relations with positive frequencies in the pressure range of interest (Fig. 4). This behaviour is analogous to ice X, which has only real positive phonon frequencies in the regime in which the classical limit predicts symmetrization of the hydrogen bond^{25–28}. The anharmonic renormalization of the phonon energies is huge, especially for the H–S bond-stretching modes in which H atoms move towards the neighbouring S atoms, which are precisely those modes that drive the second-order phase transition between the $Im\bar{3}m$ and $R3m$ phases. Therefore, the proximity to the second-order quantum phase transition is the origin of the strong anharmonicity.

Although the bond symmetrization in ice X occurs in an insulating system, H_3S is metallic and the symmetrization strongly affects the superconductivity. The calculated electron–phonon coupling and T_c lend further support to the suggestion that $Im\bar{3}m$ H_3S yields the record $T_c = 203$ K. We use Wannier-interpolated electron–phonon matrix

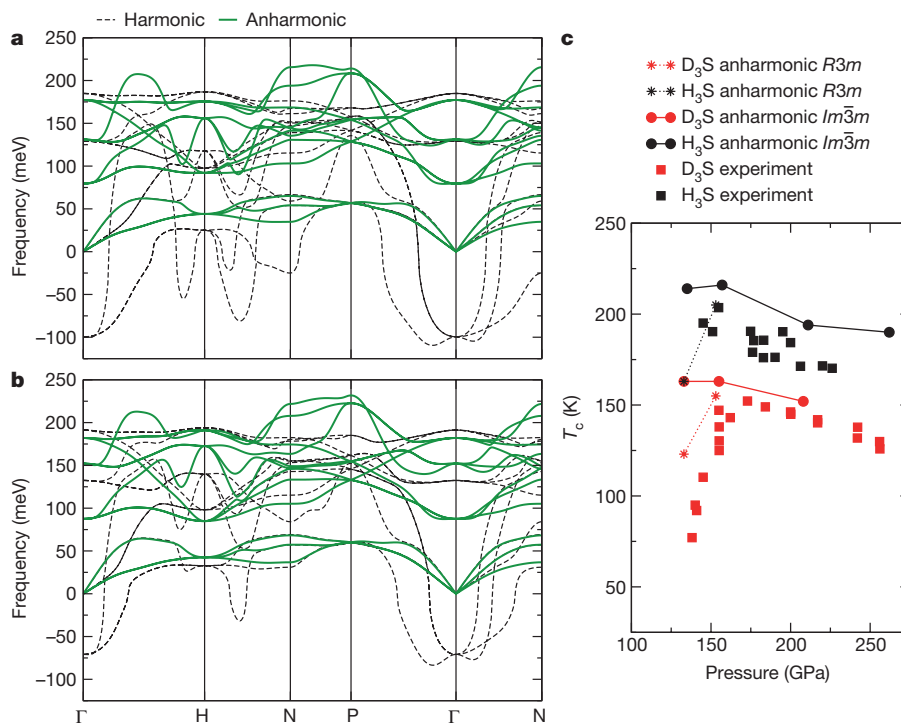


Figure 4 | Phonon spectra and superconducting transition temperature. **a**, **b**, Harmonic and SSCHA anharmonic phonon spectra of the cubic high-symmetry $Im\bar{3}m$ structure for H_3S at different pressures: 135 GPa (**a**) and 157 GPa (**b**). **c**, T_c calculated with the anharmonic phonons for the $Im\bar{3}m$

structure compared with experimental results obtained after annealing⁴. T_c results obtained with anharmonic phonons for the $R3m$ structure below 175 GPa are also shown. Each pressure takes the vibrational energy into account.

elements in our calculations²⁹ and estimate T_c by solving the isotropic Migdal–Eliashberg equations. The phonon frequencies and polarizations that enter the electron–phonon calculations are calculated using the SSCHA. Therefore, in the present treatment we neglect nonlinear corrections of the electron–phonon vertices and non-adiabatic effects that could arise from the small mass of H. The results obtained for the $Im\bar{3}m$ structure using anharmonic phonon frequencies agree well with experimental measurements of T_c for H_3S and D_3S , and correctly capture the observed increase in T_c with decreasing pressure. We also find an isotope coefficient $\alpha = -\{\ln[T_c(D_3S)] - \ln[T_c(H_3S)]\}/\ln(2)$ for $H \rightarrow D$ substitution of $\alpha = 0.35$ at 210 GPa and $\alpha = 0.40$ at 155 GPa in good agreement with experiment (see Fig. 4c). The electron–phonon coupling constant λ , which scales with the phonon frequencies as $1/\omega^2$, is enhanced with decreasing pressure owing to the overall softening of the phonon modes. This explains the smooth decrease in T_c with increasing pressure. Between approximately 130 GPa and 150 GPa, the increase in λ is compensated by the decrease in the average phonon frequency, and T_c saturates. We also present SSCHA calculations for the $R3m$ structure keeping the centroids at the $Q = 1$ position. We find a rapid decrease in T_c with decreasing pressure as in previous harmonic calculations¹³. Therefore, the observed high- T_c superconductivity cannot be explained by H_3S in the $R3m$ phase, although the sudden decrease in T_c measured for D_3S below 150 GPa (ref. 4) could be an indication of the symmetry breaking that we predicted at 115 GPa. Indeed, the predicted transition pressure depends on the choice of the exchange–correlation functional (see Methods). Even if our choice of the PBE (Perdew–Burke–Ernzerhof) exchange–correlation functional³⁰ seems appropriate, on the basis of the agreement between the experimentally observed equation of state¹⁰ and DFT calculations, we cannot exclude a small error in the transition pressure. However, the isotopic shift in the transition pressure is independent of the functional. Thus, if the decrease in T_c observed experimentally in D_3S is due to the transition from $R3m$ to $Im\bar{3}m$, then we predict that a similar decrease also occurs in H_3S , but is shifted to lower pressures by about 12 GPa. Considering that the $Im\bar{3}m$ phase has no Raman active modes, if the drop in T_c in D_3S coincides with the second-order phase transition, then we predict that Raman peaks from the $R3m$ phase would emerge at that pressure.

A recent experiment⁴ suggests that room-temperature superconductivity is reachable in other hydrogen-rich compounds. Our results show that in such hydrogen-rich materials with prospects for high- T_c superconductivity, the quantum motion of the proton induces non-trivial effects that strongly affect the thermodynamical stability, the hydrogen chemical bonding and the electron–phonon coupling strength.

Online Content Methods, along with any additional Extended Data display items and Source Data, are available in the online version of the paper; references unique to these sections appear only in the online paper.

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Author Information Reprints and permissions information is available at www.nature.com/reprints. The authors declare no competing financial interests. Readers are welcome to comment on the online version of the paper. Correspondence and requests for materials should be addressed to I.E. (ion.errea@ehu.eus), M.C. (matteo.calandra@impmc.upmc.fr) or F.M. (francesco.mauri@uniroma1.it).

METHODS

Calculation details. Supercell calculations for the SSCHA^{22,23} and linear response calculations³¹ were performed within DFT and the generalized gradient approximation (GGA) functional³⁰ as implemented in the Quantum ESPRESSO³² code. We used ultrasoft pseudopotentials³³, a plane-wave cut-off energy of 60 Ry for the kinetic energy and 600 Ry for the charge density. The charge density and dynamical matrices were calculated using a $32 \times 32 \times 32$ Monkhorst–Pack-shifted electron–momentum grid for the unit cell calculations. This mesh was adjusted accordingly in the supercell calculations. The electron–phonon coupling was calculated by using electron and phonon momentum grids composed of up to $42 \times 42 \times 42$ randomly displaced points in the Brillouin zone. The isotropic Migdal–Eliashberg equations were solved using 512 Matsubara frequencies and a Coulomb pseudopotential $\mu^* = 0.16$.

The SSCHA calculations were performed using a $3 \times 3 \times 3$ supercell for both H_3S and D_3S in the $Im\bar{3}m$ phase, yielding dynamical matrices on a commensurate $3 \times 3 \times 3$ q -point grid. The difference between the harmonic and anharmonic dynamical matrices in the $3 \times 3 \times 3$ phonon momentum grid was interpolated to a $6 \times 6 \times 6$ grid. Upon adding the harmonic matrices to the result, the anharmonic dynamical matrices were obtained on a $6 \times 6 \times 6$ grid. These dynamical matrices were used for the anharmonic electron–phonon coupling calculation. The SSCHA calculations for $Q=1$ were performed with a $2 \times 2 \times 2$ supercell. For consistency, the vibrational energies presented in Fig. 2 were also calculated using a $2 \times 2 \times 2$ supercell. However, the electron–phonon calculations for $Q=1$ were performed with the SSCHA dynamical matrices interpolated to a $6 \times 6 \times 6$ grid from the $2 \times 2 \times 2$ mesh.

The $E_{\text{vib}}(Q)$ curves in Fig. 2 were obtained as follows. E_{vib} was calculated for $Q=0$ and $Q=1$ with the SSCHA. With the SSCHA calculation at $Q=1$, we extracted $\frac{dE_{\text{vib}}}{dQ}(Q=1)$ with no further computational effort. Considering that the derivative of the curve at $Q=0$ vanishes by symmetry, we obtain straightforwardly a potential of the form $E_{\text{vib}}(Q) = A + BQ^2 + CQ^4$ (in which A , B and C are constants). The E_{vib} fit curves presented in Fig. 2 were obtained in this way. The extra point obtained at $Q=0.5$ for H_3S at $V = 97.85a_0^3$ (see Fig. 2a) confirmed the validity of the fitting procedure. The $E_{\text{BO}}(Q)$ BOES energies were calculated for many Q points, yielding an accurate fitting curve. Figure 3 was obtained using a polynomial interpolation of the BOES in the volume range shown and adding the $E_{\text{vib}}^{R3m}(x) - E_{\text{vib}}^{Im\bar{3}m}$ curves that are practically independent of volume.

Symmetry analysis of the second-order phase transition. The bond-symmetrization transition from $R3m$ to $Im\bar{3}m$ is a second-order transition and is driven by the softening of an optical mode at Γ . The mode driving the transition belongs to the irreducible representation T_{1u} , also denoted as Γ_4^- , whose dimension is three (refs 34, 35). This irreducible representation is compatible with a group–subgroup relation between the $Im\bar{3}m$ and $R3m$ space groups. Because the mode driving the transition is at the Γ point, the transition occurs without increasing the size of the unit cell, which contains four atoms in the primitive cell of the bcc lattice of the $Im\bar{3}m$ structure as well as in the rhombohedral lattice of the $R3m$ phase.

In a transition from the $m\bar{3}m(O_h)$ point group to the $3m(C_{3v})$, the most general free energy expansion contains only even terms of the order parameters³⁶. Therefore, according to Landau theory¹¹, the transition must be second-order. Our reaction coordinate Q used to describe the second-order transition is a simplified order parameter. The fact that $E(Q) = E(-Q)$ confirms that the transition is second-order.

Equations of state. In Extended Data Fig. 1 we present the equation of state both for the $Im\bar{3}m$ and $R3m$ phases. The pressure is calculated both with and without the vibrational contributions to the energies of H_3S and D_3S . The curves that include the vibrational contribution to the energy are used to determine the vibrational effects on the calculated pressures. The vibrational energy calculated includes anharmonicity in all cases as calculated by the SSCHA. The vibrational energy has a small contribution to the calculated pressure, which varies slightly with the isotopic mass. The pressure–volume curve can be efficiently fitted to the third-order Birch–Murnaghan equation. The parameters of the fit are given in Extended Data Table 1.

Dependence on the exchange–correlation functional. All calculations presented here are calculated within the GGA under the PBE parameterization³⁰. However, we performed additional calculations within the local density approximation (LDA)³⁷ and the BLYP (Becke–Lee–Yang–Parr)^{38,39} parameterization of the GGA. The second-order phase transition at the static level from $Im\bar{3}m$ to $R3m$, which

occurs at 175 GPa within PBE, is reduced to 145 GPa in the LDA and increased to 257 GPa with the BLYP parameterization. In Extended Data Fig. 2 we compare the relative coordinate $x = (d_2 - a/2)/(a/2)$ as a function of volume, where d_2 is the length of the hydrogen bond and a is the lattice parameter. The x coordinate is obtained both with the static BOES energy E_{BO} and with the total energy $E = E_{\text{BO}} + E_{\text{vib}}$. However, the vibrational energy is assumed to be independent of the functional, and the value obtained with PBE is used for estimating E_{vib} for both LDA and BLYP. We believe that this assumption is justified because the differences between the vibrational energies of $Im\bar{3}m$ and $R3m$ calculated within the SSCHA and using PBE are very weakly volume dependent, as shown in Extended Data Fig. 3. At the static level, the LDA symmetrizes the structure at a pressure 30 GPa below that predicted by PBE, whereas BLYP symmetrizes it at a pressure 82 GPa above that predicted by PBE. When including the vibrational contribution as described above, the transition between $Im\bar{3}m$ and $R3m$ occurs at 170 GPa for H_3S and 185 GPa for D_3S within BLYP. The shift in the transition pressure induced by the ZPE and its isotope dependence is similar for PBE and BLYP. The fully symmetric $Im\bar{3}m$ H_3S structure is always favoured within the LDA for H_3S over the pressure range studied, and it only becomes unfavourable below 89 GPa for D_3S .

Although the choice of exchange–correlation functional adds some uncertainty to the predicted transition pressure, we believe PBE is the most appropriate choice. This statement is based on the fact that PBE best reproduces the experimental equation of state results of ref. 10. BLYP overestimates the volume, especially when the vibrational contribution is considered (see Extended Data Fig. 4). Indeed, it is well known that BLYP vastly overestimates the equilibrium volumes in metals⁴⁰ and, therefore, it is not the best choice for H_3S . On the contrary, when including the vibrational contribution to the pressure, which is calculated in all cases using the PBE functional, the PBE equation of state is in rather good agreement with experimental results¹⁰ and certainly performs better than LDA and BLYP.

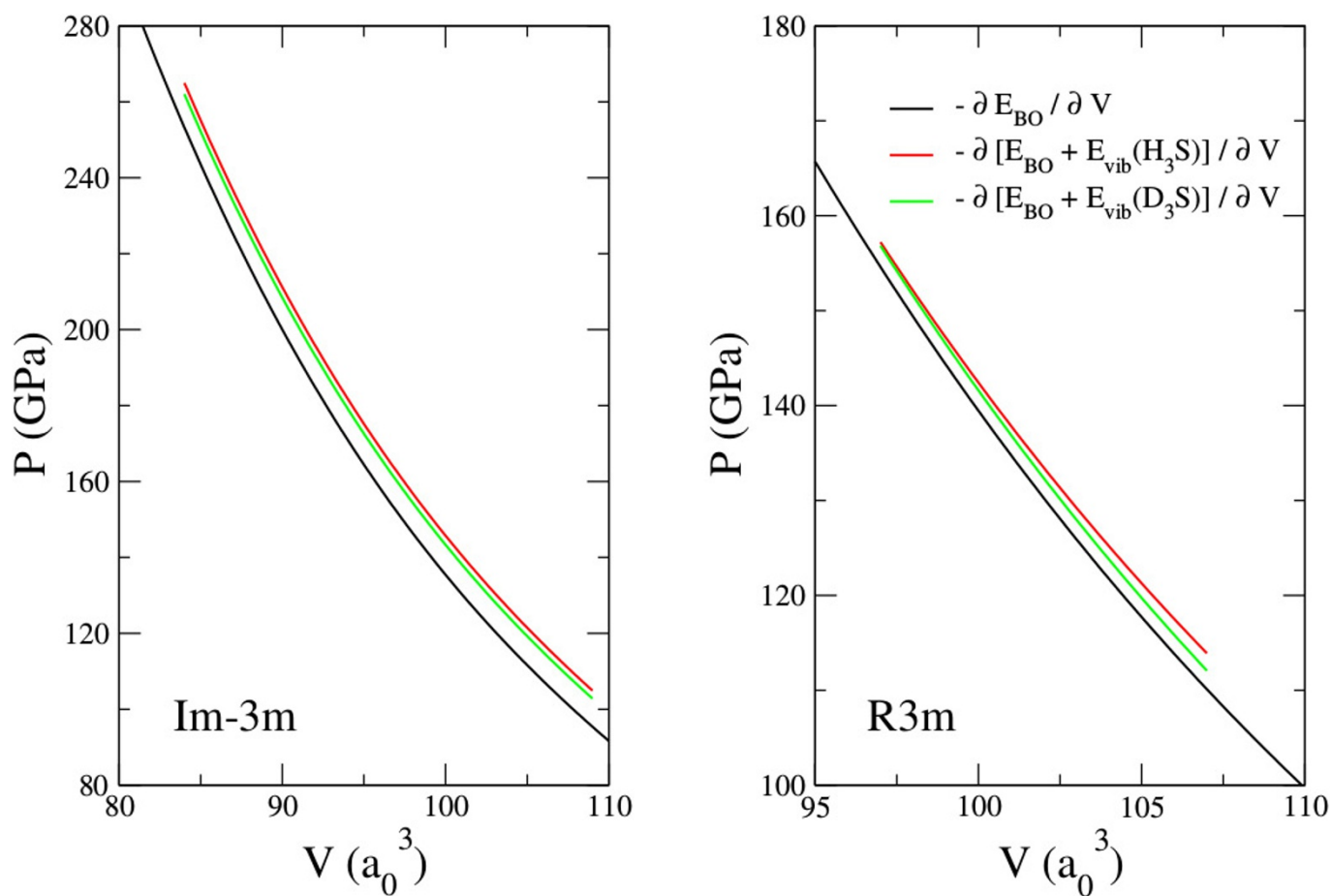
Phonons of $Im\bar{3}m$ D_3S . In Extended Data Fig. 5 we show the harmonic and anharmonic phonon spectra of D_3S in the $Im\bar{3}m$ phase at two different pressures.

Anharmonic phonon dispersion in the $R3m$ phase. In Extended Data Fig. 6 we show the SSCHA phonon spectra calculated for the $R3m$ phase at two different pressures. Here the centroid position of the SSCHA is placed at the minimum of the BOES, which is not a minimum of the total energy, as discussed. The Raman active modes of the $R3m$ phase are listed in Extended Data Table 2.

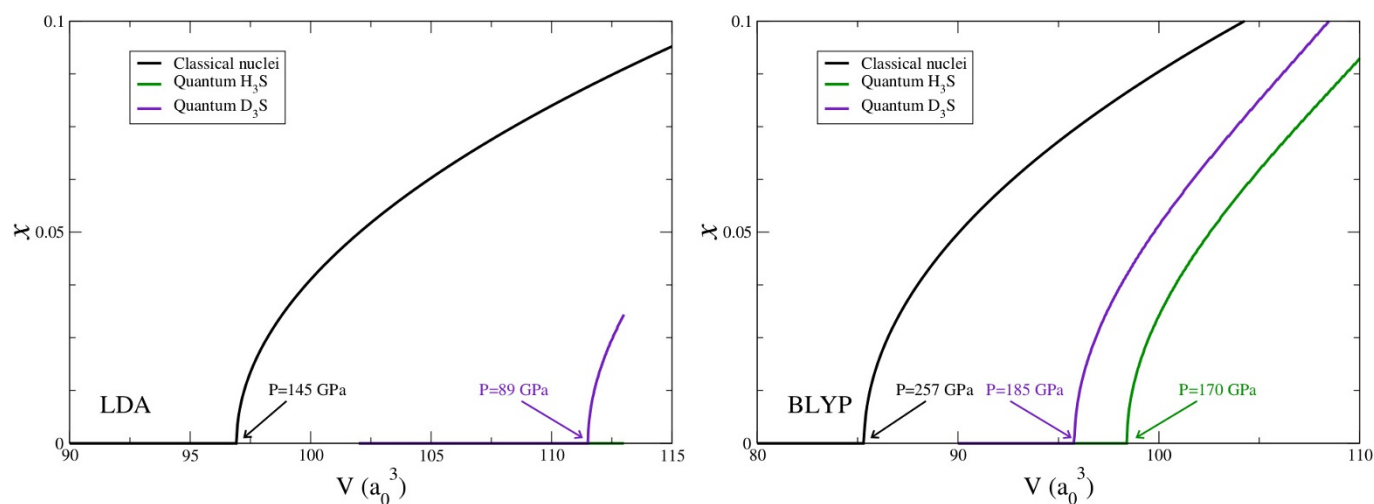
Superconducting properties. In Extended Data Table 3 we summarize the calculated values of the electron–phonon coupling constant λ , logarithmic frequency average ω_{log} and T_c .

In Extended Data Fig. 7 we show the Eliashberg function $\alpha^2F(\omega)$ and integrated electron–phonon coupling constant $\lambda(\omega)$ of the $Im\bar{3}m$ phase at two different pressures calculated using the SSCHA method. The Eliashberg function $\alpha^2F(\omega)$ in the anharmonic case for the $R3m$ structure is given in Extended Data Fig. 6.

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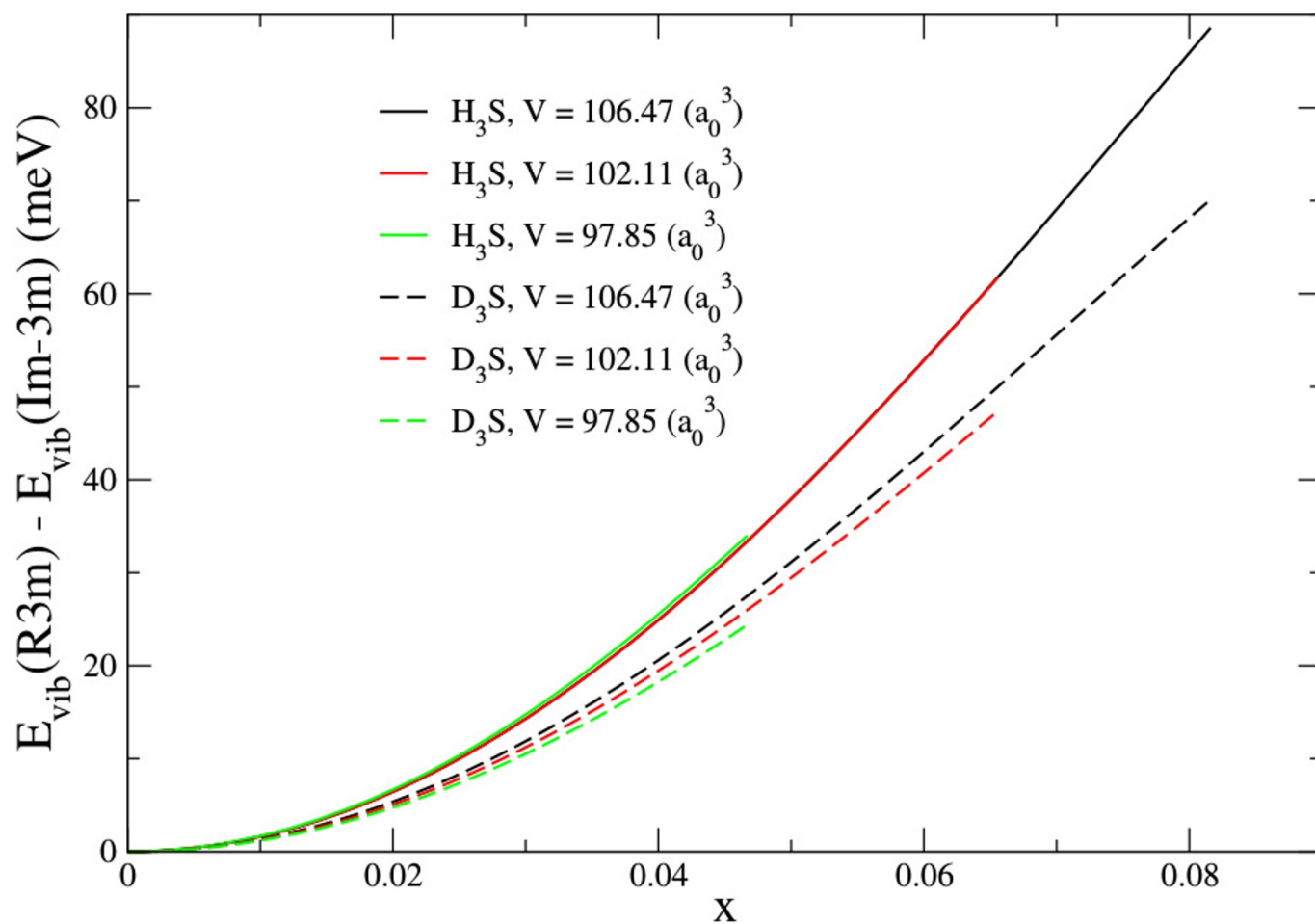


Extended Data Figure 1 | Equations of state. Pressure P as a function of volume V for the *Im-3m* (left) and *R3m* (right) phases, calculated from the static energy E_{BO} that represents the classical nuclei limit (black), and including the vibrational contribution within the SSCHA for both H_3S (red) and D_3S (green).

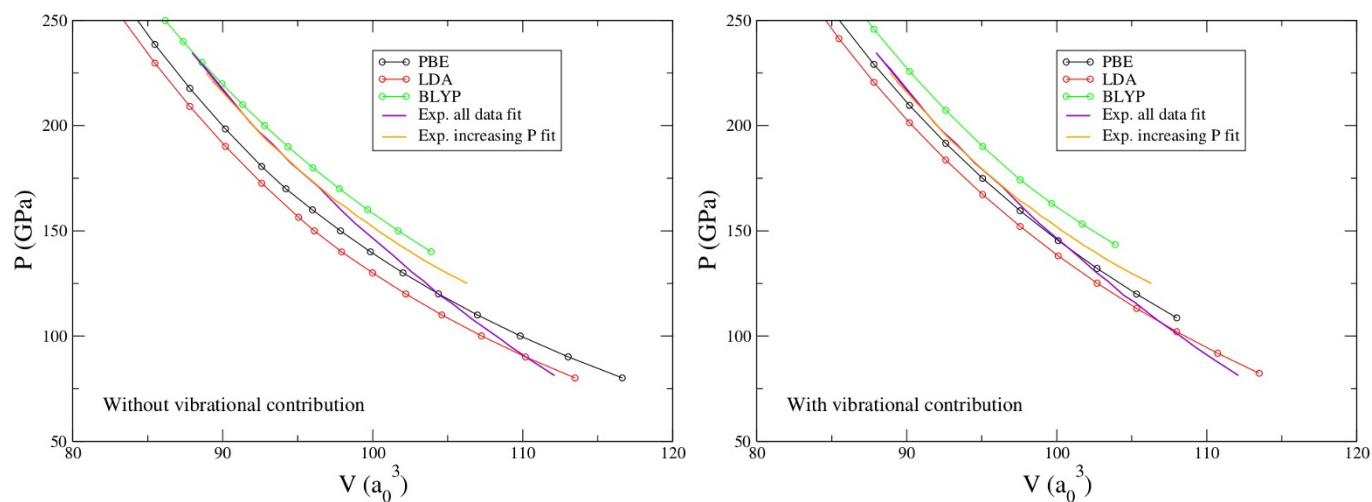


Extended Data Figure 2 | Bond symmetrization within LDA (left) and BLYP (right). For each volume V the relative coordinate $x = (d_2 - a/2)/(a/2)$, in which d_2 is the length of the hydrogen bond and a is the lattice parameter, obtained at the energy minimum is given. When $x = 0$, the covalent and hydrogen bonds have the same length and the structure is

fully symmetric. The energy is calculated at the static level without any vibrational contribution as derived from the BOES (black), and including the quantum anharmonic vibrational contribution both for H_3S (green) and D_3S (purple). The pressure P below which the cubic structure distorts is given in each case.

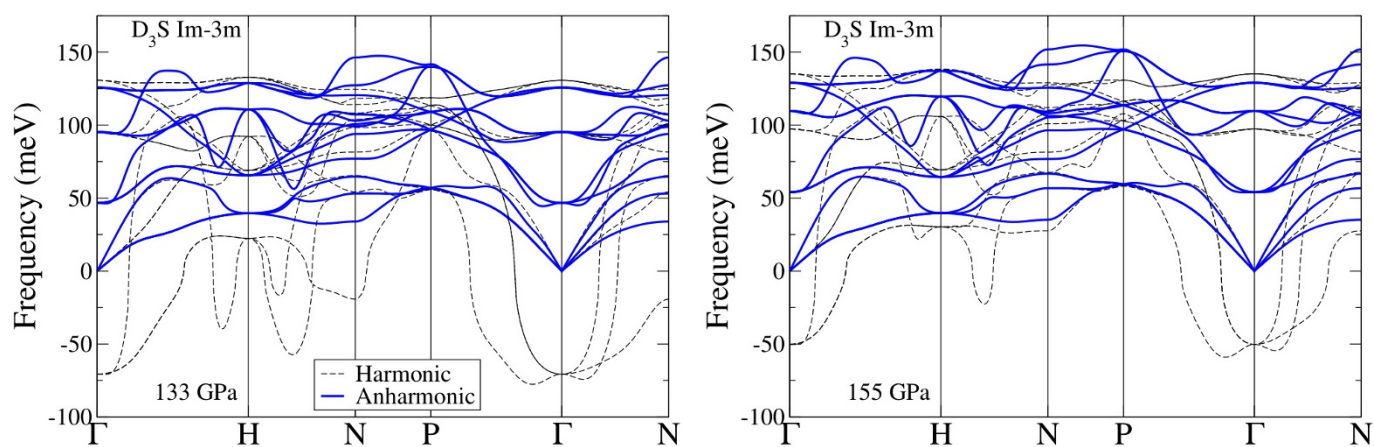


Extended Data Figure 3 | Vibrational energy. SSCHA anharmonic vibrational energy ($E_{\text{vib}}^{\text{R3m}}(x) - E_{\text{vib}}^{\text{Im3m}}$) calculated as a function of the relative coordinate $x = (d_2 - a/2)/(a/2)$ for different volumes (see legend).

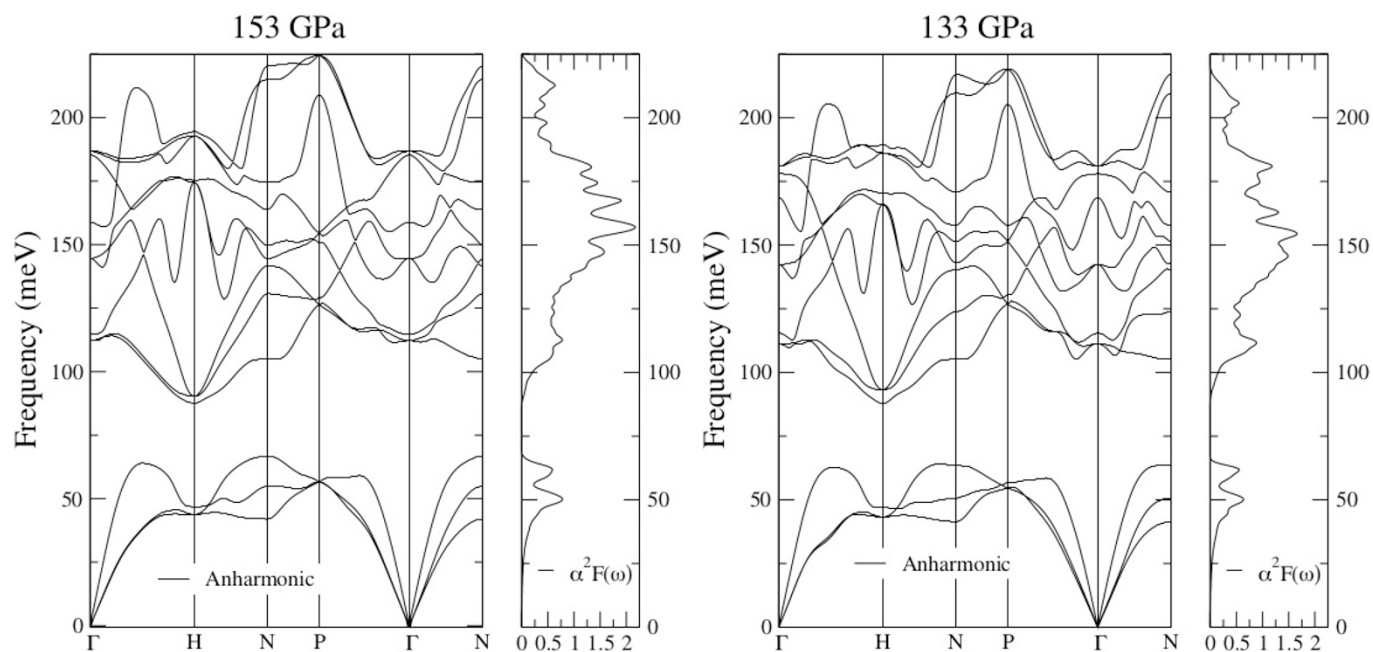


Extended Data Figure 4 | Dependence of the equation of state on the density functional. The equation of state is calculated with different exchange correlation functionals with (right) and without (left) the vibrational contribution to the pressure. At each volume V the pressure P

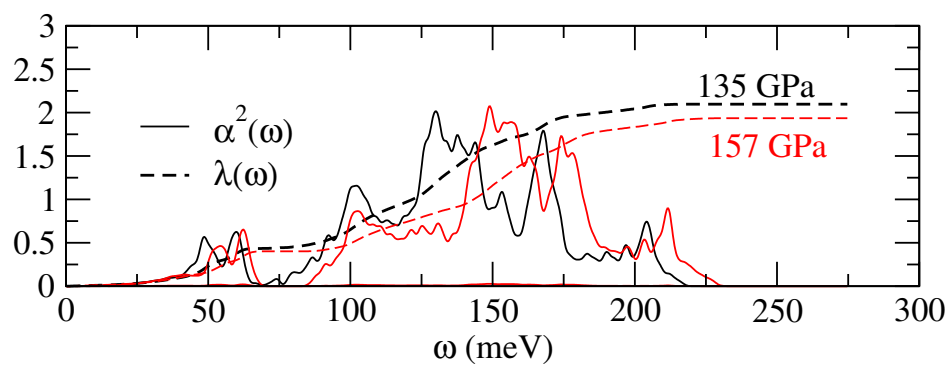
of the structure with minimum energy is given, which depends on whether or not the vibrational contribution is included (see Extended Data Fig. 2). The results are compared with the two curves obtained experimentally¹⁰.



Extended Data Figure 5 | Phonons of $Im\bar{3}m$ D_3S . Comparison between the harmonic and anharmonic phonons of $Im\bar{3}m$ at two different pressures for D_3S : 133 GPa (left) and 155 GPa (right).



Extended Data Figure 6 | Anharmonic phonons of $R3m$ H_3S . Anharmonic phonons are shown at two different pressures: 153 GPa (left) and 133 GPa (right). The Eliashberg functions $\alpha^2F(\omega)$ are also shown.



Extended Data Figure 7 | Superconducting properties of $Im\bar{3}m$ H_3S . Anharmonic Eliashberg function $\alpha^2F(\omega)$ (solid lines) and integrated electron–phonon coupling constant $\lambda(\omega)$ (dashed lines) of the $Im\bar{3}m$ phase as functions of frequency ω at two different pressures: 135 GPa (black) and 157 GPa (red).

Extended Data Table 1 | Birch–Murnaghan fit to the equation of state

	B_0 (GPa)	V_0 (a_0^3)	B_0'
Classical nuclei	129.8	158.4	3.6
H ₃ S	86.63	176.4	3.9
D ₃ S	87.81	174.8	3.9

Parameters of the fit to $P(V) = \frac{3}{2}B_0[(V_0/V)^{7/3} - (V_0/V)^{5/3}]\{1 + \frac{3}{4}(B_0' - 4)[(V_0/V)^{2/3} - 1]\}$ for the equation of state of the $Im\bar{3}m$ phase with classical nuclei, and with vibrational contributions for both H₃S and D₃S. B_0 , bulk modulus at zero pressure; V_0 , reference volume; B_0' , pressure derivative of the bulk modulus at zero pressure.

Extended Data Table 2 | Raman and infrared active modes

<i>Im3m</i>							
Mode	Degeneracy	R or I active?	ω (meV)				
			H ₃ S		D ₃ S		
			135 GPa	157 GPa	133 GPa	155 GPa	
T_{1u}	3	I	79.6	87.6	46.7	54.1	
T_{1u}	3	I	131.0	152.1	95.4	109.8	
T_{2u}	3		177.3	182.0	125.8	129.0	
<i>R3m</i>							
Mode	Degeneracy	R or I active?	ω (meV)				
			H ₃ S		D ₃ S		
			133 GPa	153 GPa	133 GPa	153 GPa	
E	2	R+I	111.3	112.2	74.4	73.7	
A_1	1	R+I	115.4	114.7	79.8	73.7	
E	2	R+I	142.4	144.6	105.6	105.9	
A_1	1	R+I	168.7	158.8	123.2	115.0	
A_2	1		178.0	185.3	125.1	130.7	
E	2	R+I	181.0	186.7	127.0	131.7	

Phonon modes at the Γ point for the *Im $\bar{3}m$* and *R3m* phases at different pressures calculated including anharmonicity within the SSCHA. The results for *R3m* are obtained keeping the centroid position at the minimum of the BOES. Raman (R) and infrared (I) activity is indicated for each mode.

Extended Data Table 3 | Superconducting parameters

	Compound	P (GPa)	λ	$\omega_{\log}$ (meV)	T_c (K)
SSCHA $Im\bar{3}m$	H ₃ S	135	2.10	104.9	214
	H ₃ S	157	1.94	113.8	216
	D ₃ S	133	2.45	70.5	163
	D ₃ S	155	2.08	82.9	163
SSCHA $R\bar{3}m$	H ₃ S	133	1.62	105.8	163
	H ₃ S	153	1.93	109.3	205
	D ₃ S	132	1.64	79.8	123
	D ₃ S	152	2.00	81.4	155

Calculated λ , $\omega_{\log}$ and T_c . The pressure given includes the vibrational contribution.